

Solutions to Exercise Sheet: Networking Fundamentals

Exercise 1

1.1: Describes the term “network edge” in your own words.

It is where the hosts are located.

1.2: Describes the term “access network” in your own words.

It is how hosts become interconnected and get access to other networks.

1.3: Describes the term “network core” in your own words.

It is a mesh of interconnected routers.

1.4: Which layer is the uppermost in the current Internet protocol stack?

The Application Layer is the uppermost layer in the current Internet protocol stack.

1.5: Describe the differences between packet and circuit switching.

Packet Switching	Circuit Switching
Data is divided into packets that are transmitted independently; packets share resources dynamically, good for bursty traffic; may have loss and delay.	Dedicated path allocated end-to-end; reliable, no packet loss due to congestion.

The fundamental difference between packet and circuit switching lies in resource allocation and connection establishment: circuit switching creates a dedicated, reserved path with guaranteed bandwidth for the entire communication session (like traditional phone calls), ensuring reliable performance but potentially wasting resources during idle periods, while packet switching breaks data into independent packets that share network resources dynamically without requiring a pre-established connection, making it more efficient for bursty data traffic but with variable delays and potential packet loss during congestion.

1.6: How are ISPs interconnected?

ISPs are interconnected via tier-1 ISPs, regional ISPs, and Internet Exchange Points (IXPs) where peering links connect them.

Also an accepted solution:

ISPs are interconnected through a hierarchical structure:

- **Tier 1 ISPs:** Large national/global ISPs that interconnect with each other
- **Regional ISPs:** Connect access ISPs to Tier 1 ISPs
- **Internet Exchange Points (IXPs):** Facilities where ISPs peer directly
- **Peering links:** Direct connections between ISPs
- **Content Delivery Networks (CDNs):** Large content providers create their own networks to get closer to users

1.7: How does packet loss occur?

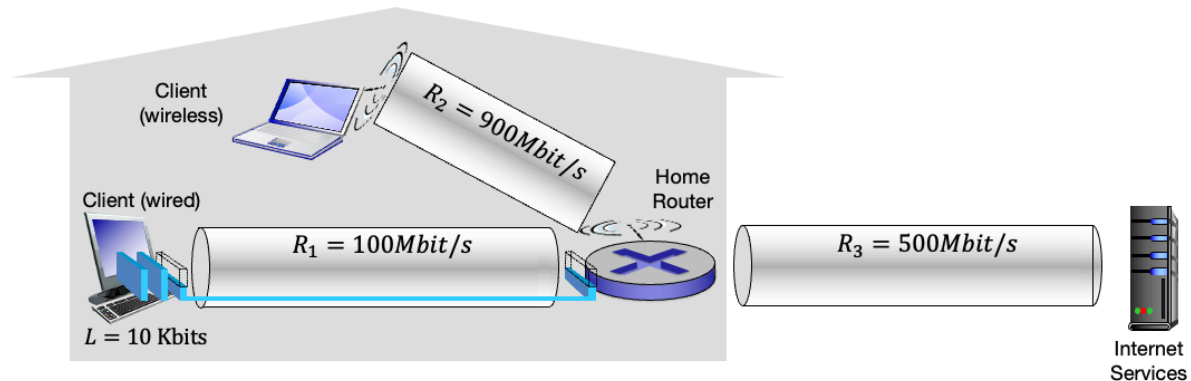
Packet loss occurs when routers' buffers fill due to congestion; excess packets are dropped because they cannot be stored on the router.

1.8: Describe Cerf and Kahn's internetworking principles and how they influence today's Internet in your own words.

Cerf and Kahn designed four main principles that still define the internet and how it works today. (1) Minimalism or autonomy means different networks can connect to the internet without having to change their internal structure, (2) the best-effort service model means the internet tries to deliver the packets but makes no promises, (3) stateless routing means router do not remember previous packets or maintain ongoing connection, each packet is handled independently based solely on its destination address, (4) decentralized control means no single organization or government runs the internet, its distributed with everyone managing their own pieces while following standard protocols to work together. These principles explain why the Internet is so resilient and scalable, but also why it initially had limited security features.

Exercise 2

Assume the network topology as shown below.



2.1: What are the bottlenecks in this network for the connection between (a) the wired client and the Internet Services, (b) the wireless client and the Internet Services, (c) the wired and the wireless client?

(a) Between the wired Client and the Internet Services: $R_1 = 100 \text{ Mbit/s}$

(b) Between the wireless Client and the Internet Services: $R_3 = 500 \text{ Mbit/s}$

(c) Between the wired Client and the wireless Client: $R_1 = 100 \text{ Mbit/s}$

2.2: What is the packet transmission delay for connection R_1 .

L: packets length (bits) = 10 Kbits

R: link transmission rate (bps) = 100 Mbits/s

$$\text{Transmission delay} = \frac{L}{R} = \frac{10 \text{ Kbits}}{100 \text{ Mbit/s}} = 0.1 \text{ ms}$$

2.3: What is the nodal delay for the wired client, assuming the physical link R_1 has a length of 10m and the queueing delay is negligibly small.

D: length of physical link

S: propagation speed ($\sim 2 \times 10^8 \text{ m/sec}$)

$$\text{Propagation delay} = \frac{d}{S} = \frac{10 \text{ m}}{2 \times 10^8 \text{ m/s}} = 3.3 \times 10^{-8} \frac{\text{s}}{\text{m}}$$

To calculate the nodal delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

$$d_{\text{nodal}} = 0.1 \text{ ms} + 0.00033 \text{ ms} = 0.10033 \text{ ms}$$

2.4: How do the bottlenecks for (a), (b), and (c) in 2.1 change if the home router is replaced and R_1 has now a bandwidth of 1000Mbit/s?

(a) Between the wired Client and the Internet Services: $R_3 = 500$ Mbit/s

(b) Between the wireless Client and the Internet Services: no change

(c) Between the wired Client and the wireless Client: $R_2 = 900$ Mbit/s

Exercise 3

3.1: For what does a network protocol specify rules?

- **How to format messages**
- **Which actions should be taken when messages are received, or other events take place.**

3.2: Why are network protocols organized in layers?

Network protocols are organized in layers to simplify the design, implementation and management of complex network systems. The different layers take care of specific tasks and hand off to the next layer for the next task. Modularization eases maintenance and updating of the system.

3.3: Explain the term “Encapsulation” in a few sentences.

Encapsulation in networking can be termed as or referred to as a process of wrapping (or unwrapping) data with the necessary layered protocol information, i.e., the header, before it is handed to the next layer.

Exercise 4

4.1: Explain the term “Forwarding” in a few sentences.

Forwarding is the local action taken by network devices to move incoming (arriving) data packets from one interface (router’s input link) to the appropriate outgoing (output) link. Also known as Switching.

4.2: Explain the term “Routing” in a few sentences.

Routing is the process of determining the optimal (as per to be specified parameters such as speed, cost, etc.) paths for packets to travel from their source to their destination across a network. This is a global action.

Exercise 5

5.1: Explain the term “Packet Sniffing” in a few sentences.

Packet Sniffing refers to intercepting and/or monitoring packets in a broadcast medium by an attacker. A successful sniffing attack gives the attacker access to the information communicated in the packets.

5.2: Explain the term “IP Spoofing” in a few sentences.

IP Spoofing refers to an attacker injecting the packets with falsified address data. BY spoofing the origin address, an attacker can mask their own address or direct response messages to an unsuspecting host.

5.3: Explain the term “Denial of Service” in a few sentences.

When attackers attempt to make a resource (e.g., web service) unavailable by flooding the service with requests. These requests can for example aim to saturate the network link or keep the recipient busy with computations.

5.4: Which categories of the STRIDE model from the Cybersecurity Fundamentals lecture does each of the three terms above correspond?

Packet Sniffing: This corresponds to Information disclosure (I) in STRIDE

IP Spoofing: This corresponds to Spoofing (S) in STRIDE

Denial of Services: This corresponds directly to Denial of service (D) in STRIDE